

Question 1 – Line Chart (Monthly Sales)

R

```
# Monthly Sales Data
```

```
month <- c("January", "February", "March", "April", "May")
```

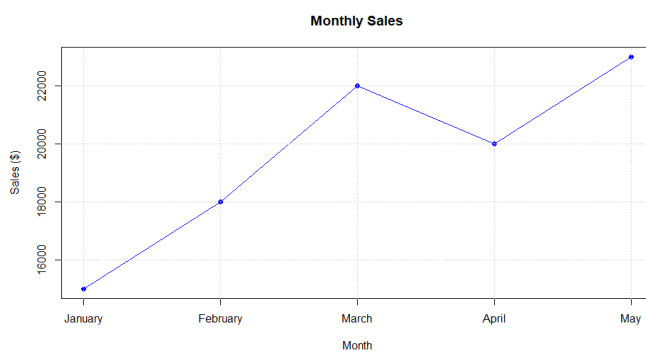
```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
# Line Chart
```

```
plot(sales,  
     type = "o",  
     pch = 19,  
     xaxt = "n",  
     col = "blue",  
     xlab = "Month",  
     ylab = "Sales ($)",  
     main = "Monthly Sales")
```

```
axis(1, at = 1:5, labels = month)
```

```
grid()
```



Question 2 – Bar Chart (Top-Selling Products)

(Since product data is not provided, sample values are used.)

R

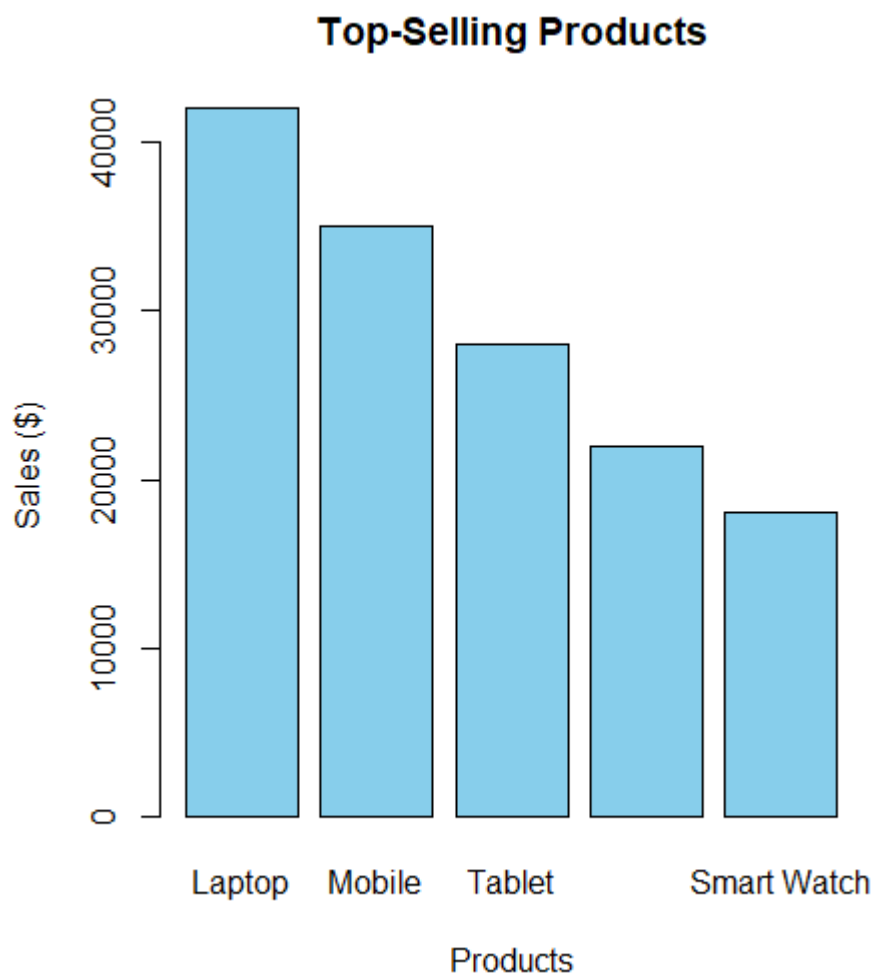
```
# Product Sales
```

```
product <- c("Laptop", "Mobile", "Tablet", "Headphones", "Smart Watch")
```

```
product_sales <- c(42000, 35000, 28000, 22000, 18000)
```

```
# Bar Chart
```

```
barplot(product_sales,  
        names.arg = product,  
        col = "skyblue",  
        main = "Top-Selling Products",  
        xlab = "Products",  
        ylab = "Sales ($)")
```



Question 3 – Scatter Plot (Advertising Budget vs Monthly Sales)

R

Advertising Budget

```
advertising <- c(3000, 4000, 5000, 4500, 5500)
```

Scatter Plot

```
plot(advertising,  
     sales,  
     pch = 19,  
     col = "red",  
     xlab = "Advertising Budget ($)",  
     ylab = "Monthly Sales ($)",  
     main = "Advertising Budget vs Monthly Sales")
```

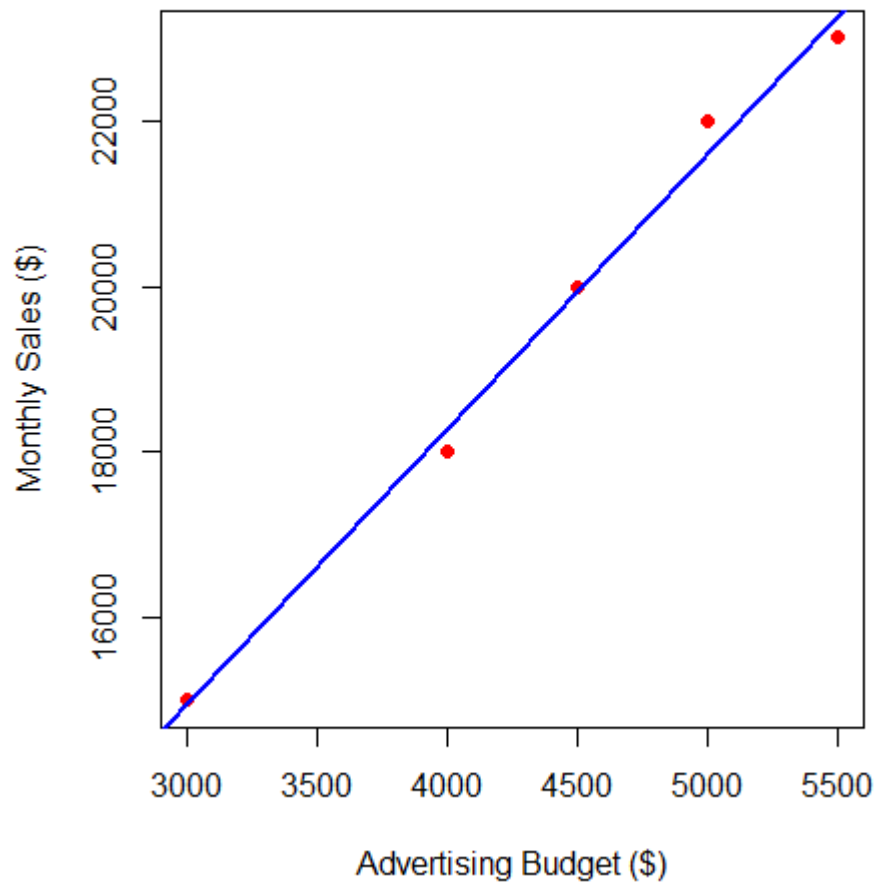
Regression Line

```
abline(lm(sales ~ advertising), col = "blue", lwd = 2)
```

Correlation

```
cor(advertising, sales)
```

Advertising Budget vs Monthly Sales



Question 4 – Interactive Dashboard

Install the required packages:

R

```
install.packages("flexdashboard")
```

```
install.packages("plotly")
```

R

```
library(plotly)
```

Interactive Line Chart

```
line_chart <- plot_ly(  
  x = month,  
  y = sales,  
  type = "scatter",  
  mode = "lines+markers"  
)
```

Product Data

```
product <- c("Laptop", "Mobile", "Tablet", "Headphones", "Smart Watch")
```

```
product_sales <- c(42000, 35000, 28000, 22000, 18000)
```

```
# Interactive Bar Chart
bar_chart <- plot_ly(
  x = product,
  y = product_sales,
  type = "bar"
)
```

```
# Display Charts
line_chart
bar_chart
```

